Topic 21

Alternating-Current Circuits and Electromagnetic Waves

Reading Question 21-1 (1 of 2)

In an AC resistive circuit, when the current is at a minimum, the potential difference is

1. at a maximum.
2. zero.
3. at a minimum.
4. impossible to tell without knowing their phase.

Reading Question 21-1 (2 of 2)

In an AC resistive circuit, when the current is at a minimum, the potential difference is

1. at a maximum.
2. **Answer: zero.**
3. at a minimum.
4. impossible to tell without knowing their phase.

Reading Question 21-2 (1 of 2)

An inductor with inductance *L* is in series with an AC source. The most effective way to increase the reactance caused by the inductor would be to

1. increase the frequency of the AC source.
2. decrease the current, I_{max}.
3. decrease the frequency of the AC source.
4. increase the potential difference, ΔV_{max}.

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Reading Question 21-2 (2 of 2)

An inductor with inductance L is in series with an AC source. The most effective way to increase the reactance caused by the inductor would be to

1. **Answer: increase the frequency of the AC source.**
2. decrease the current, I_{max}.
3. decrease the frequency of the AC source.
4. increase the potential difference, ΔV_{max}.

Reading Question 21-3 (1 of 2)

To maximize the reactance in a circuit with a capacitor and an AC source, you could

1. minimize the angular frequency and capacitance.
2. minimize the angular frequency and maximize the capacitance.
3. maximize the angular frequency and capacitance.
4. maximize the angular frequency and minimize the capacitance.

Reading Question 21-3 (2 of 2)

To maximize the reactance in a circuit with a capacitor and an AC source, you could

1. **Answer: minimize the angular frequency and capacitance.**
2. minimize the angular frequency and maximize the capacitance.
3. maximize the angular frequency and capacitance.
4. maximize the angular frequency and minimize the capacitance.

Reading Question 21-4 (1 of 2)

The impedance of an *RLC AC* circuit is dependent on

1. the resistance.
2. the resistance, inductance and capacitance.
3. the resistance and frequency.
4. the resistance, inductance, capacitance, and frequency.

Reading Question 21-4 (2 of 2)

The impedance of an *RLC AC* circuit is dependent on

1. the resistance.
2. the resistance, inductance and capacitance.
3. the resistance and frequency.
4. **Answer: the resistance, inductance, capacitance, and frequency.**

Reading Question 21-5 (1 of 2)

The primary coil in an ideal transformer has N turns. To output half the input potential difference, the secondary coil should have

1. $N / 2$ turns.
2. N turns.
3. $2N$ turns.
4. a number impossible to determine with the given information.

Reading Question 21-5 (2 of 2)

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1. **Answer: $N/2$ turns.**
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4. a number impossible to determine with the given information.

Reading Question 21-6 (1 of 2)

The electromagnetic wave intensity, *I*, is best described as

1. the energy averaged over time.
2. the rate at which energy flows through a unit surface area perpendicular to the direction of wave propagation, averaged over time.
3. the number of photons passing through a surface perpendicular to the direction of wave propagation at any given moment.
4. the concentration of waves at any given point and time.

Reading Question 21-6 (2 of 2)

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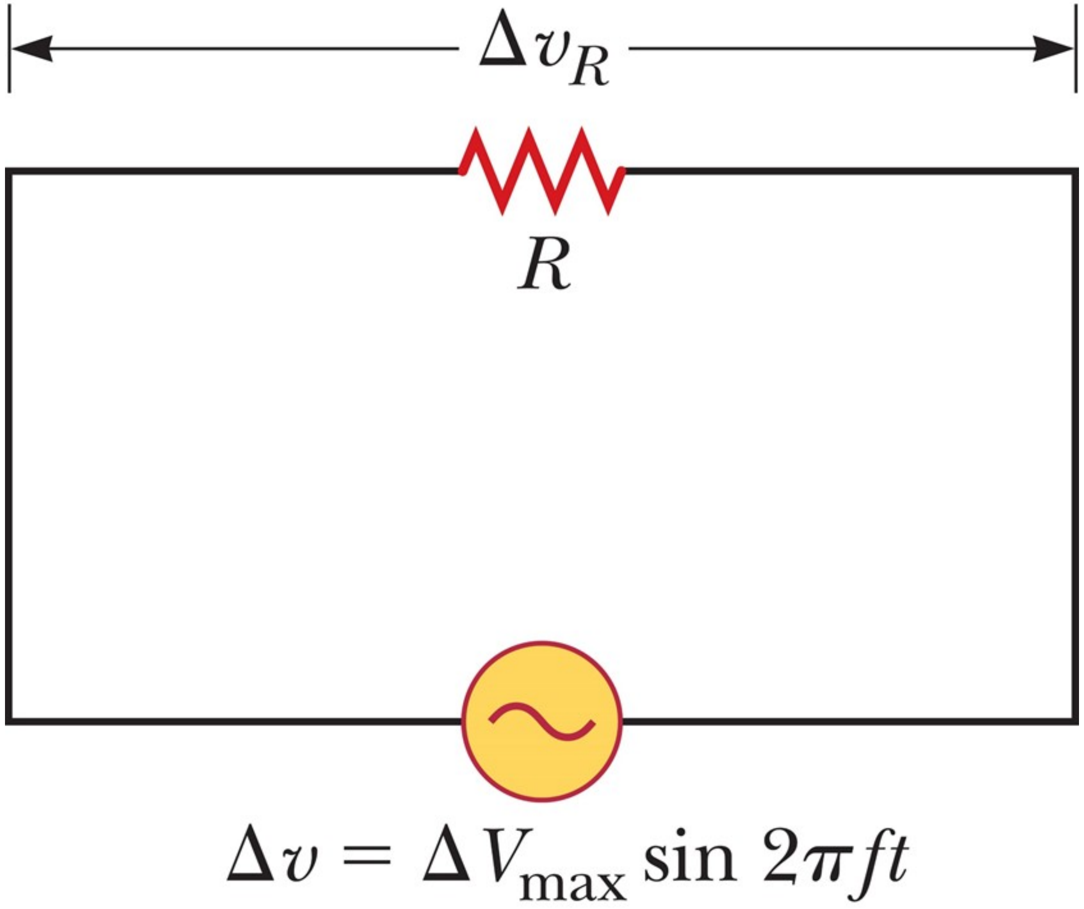
3. the number of photons passing through a surface perpendicular to the direction of wave propagation at any given moment.
4. **Answer: the concentration of waves at any given point and time.**

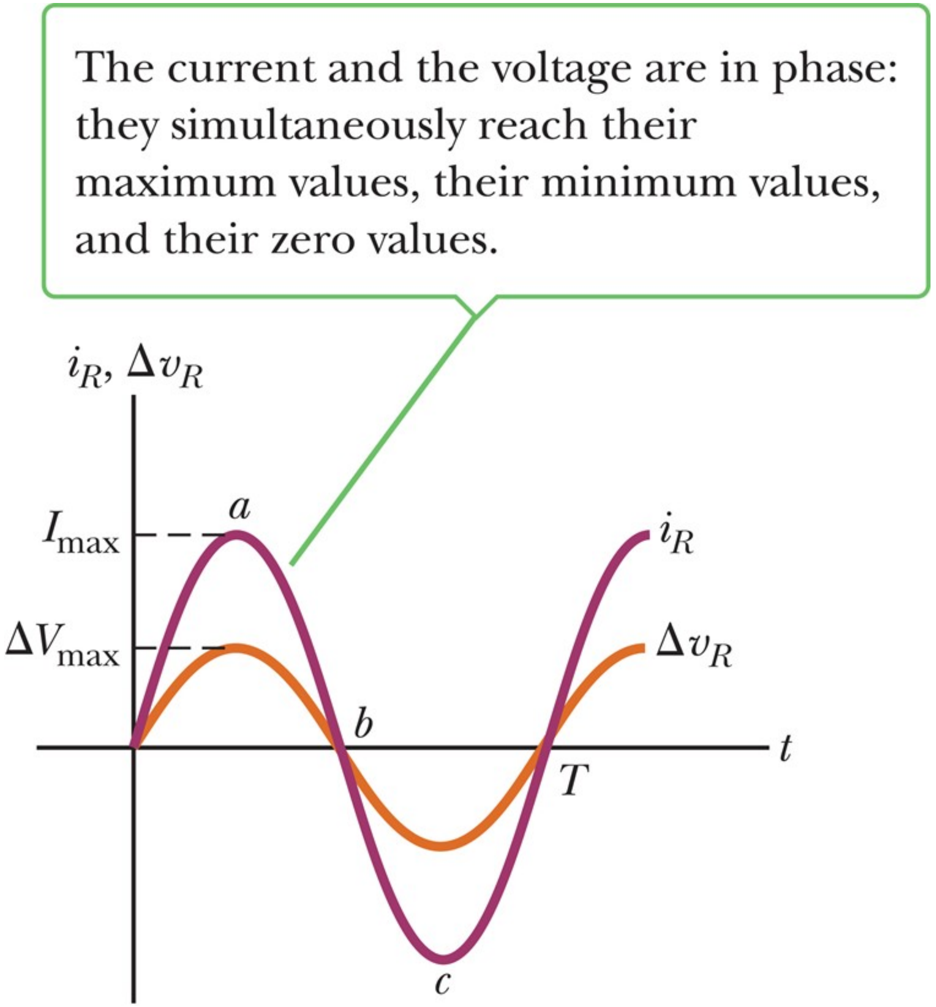
Topic 21: Alternating-Current Circuits and Electromagnetic Waves

TOPIC DISCUSSION

Resistors in an AC Circuit (1 of 5)



Resistors in an AC Circuit (2 of 5)



Resistors in an AC Circuit (3 of 5)

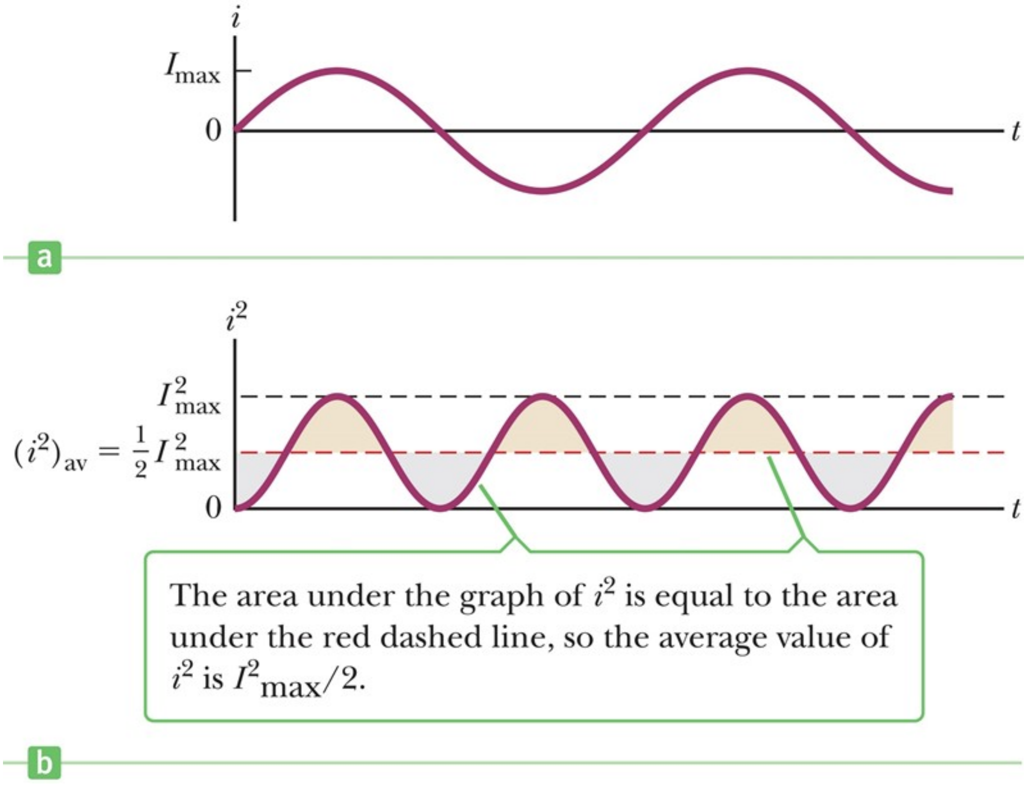
$$P = i^2 R$$

$$i^2 = \sin^2(2\pi f t)$$

$$I_{rms} = \frac{I_{max}}{\sqrt{2}} = 0.707\ I_{max}$$

$$P_{av} = I_{rms}^2 R$$

$$\Delta V_{rms} = \frac{\Delta V_{max}}{\sqrt{2}} = 0.707\ \Delta V_{max}$$

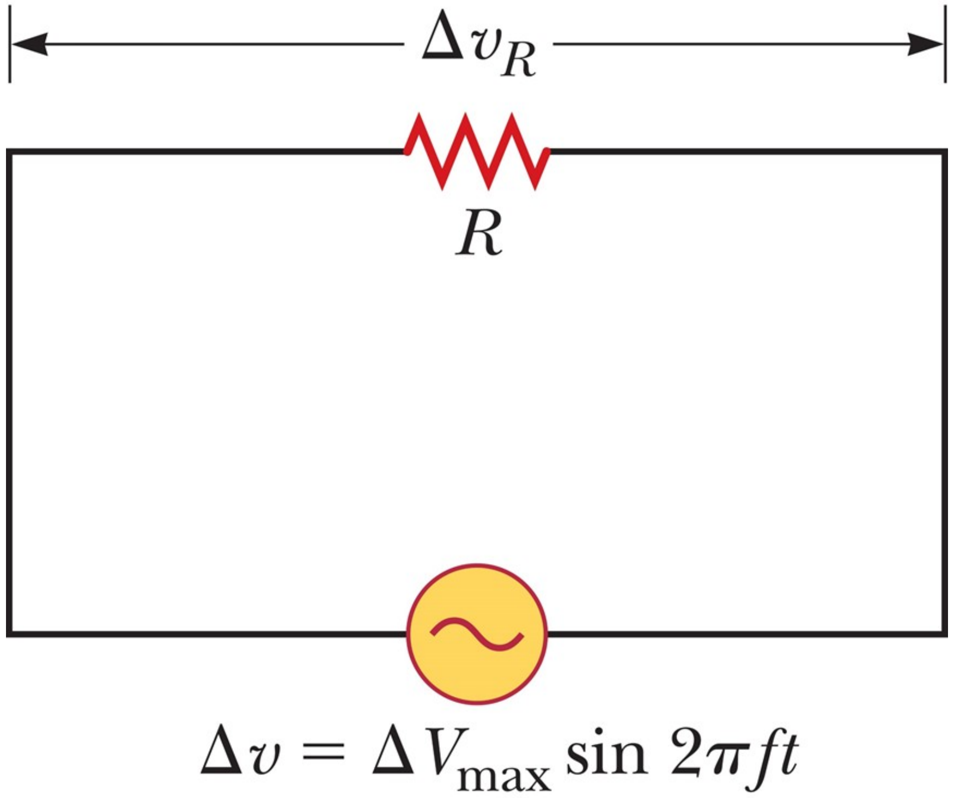


Resistors in an AC Circuit (4 of 5)

Table 21.1 Notation Used in This Topic

	Voltage	**Current**
Instantaneous value	ΔV	i
Maximum value	ΔV_{max}	I_{max}
rms value	ΔV_{rms}	I_{rms}

Resistors in an AC Circuit (5 of 5)



$$\Delta V_{\text{rms}} = I_{\text{rms}} R$$

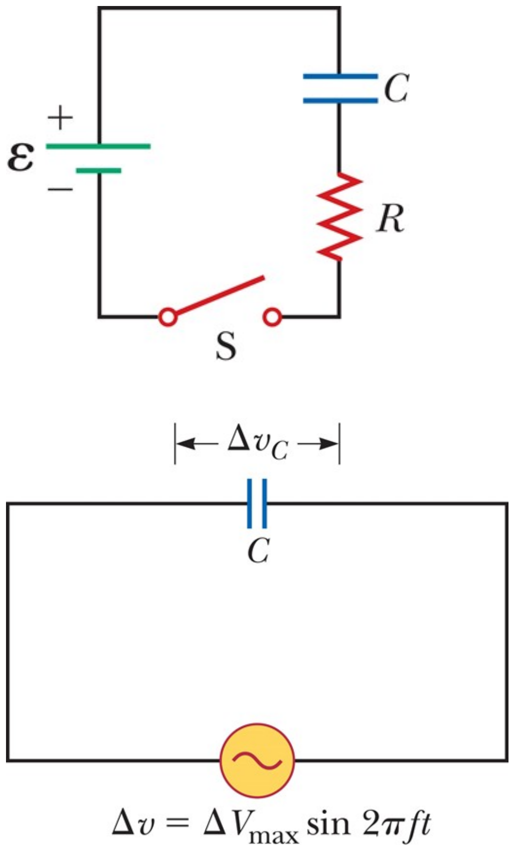
$$\Delta V_{R,\max} = I_{\max} R$$

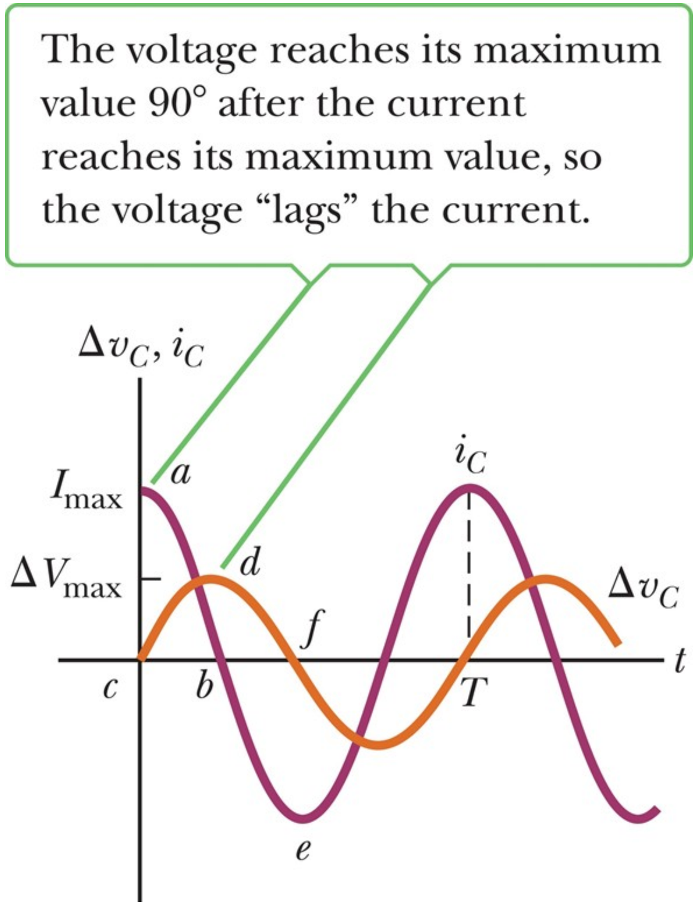
Think – Pair – Share 1

Which of the following statements can be true for a resistor connected in a simple series circuit to an operating AC generator?

1. $P_{av} = 0$ and $i_{ave} = 0$
2. $P_{av} = 0$ and $i_{ave} > 0$
3. $P_{av} > 0$ and $i_{ave} = 0$
4. $P_{av} > 0$ and $i_{ave} > 0$

Capacitors in an AC Circuit (1 of 2)





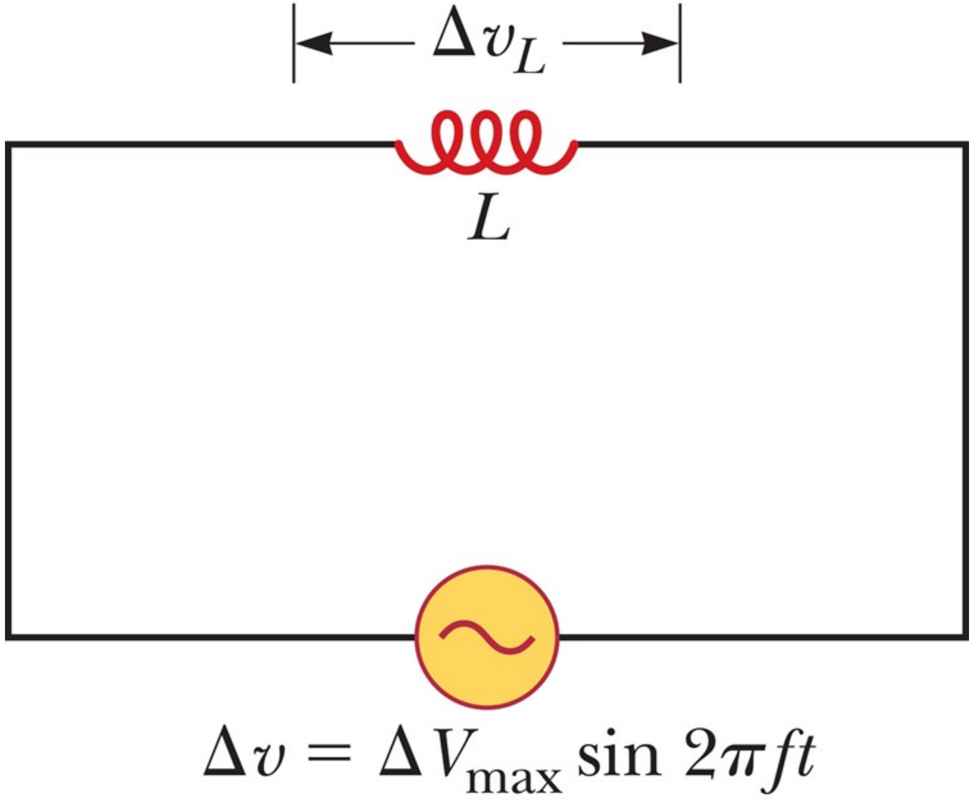
The voltage across a capacitor lags the current by 90°.

Capacitors in an AC Circuit (2 of 2)

$$X_C \equiv \frac{1}{2\pi f C} \quad \text{and} \quad \omega = 2\pi f$$

$$\Delta V_{C,\max} = I_{\text{rms}} X_C$$

Inductors in an AC Circuit (1 of 2)

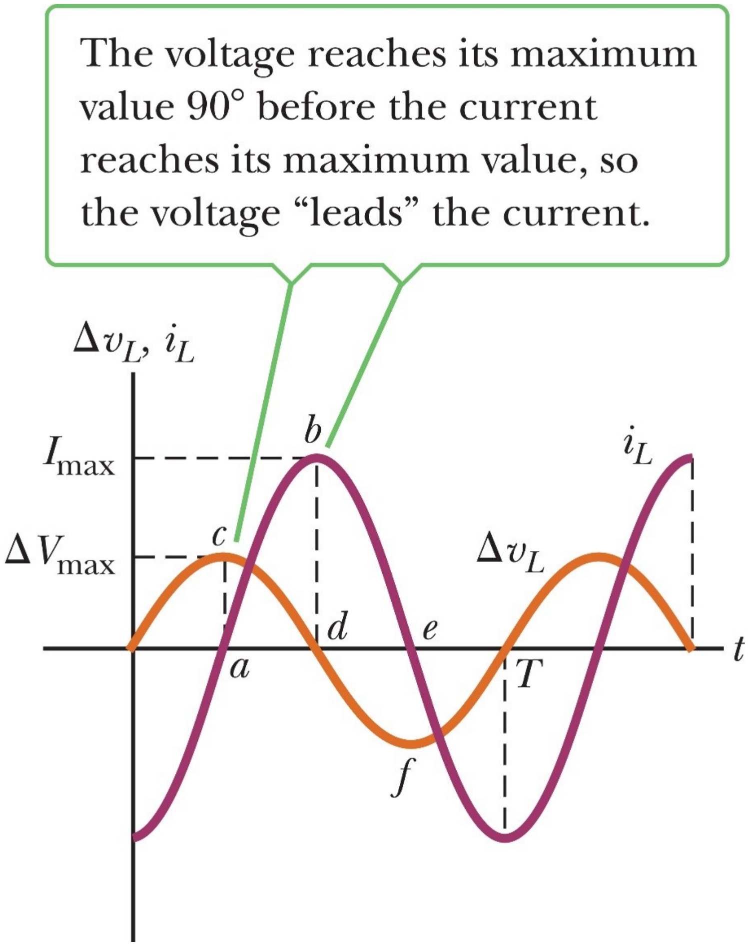


$$\Delta v_L = L\frac{\Delta I}{\Delta t}$$

$$X_L = 2\pi f L$$

$$\Delta V_{L,\text{rms}} = I_{\text{rms}} X_L$$

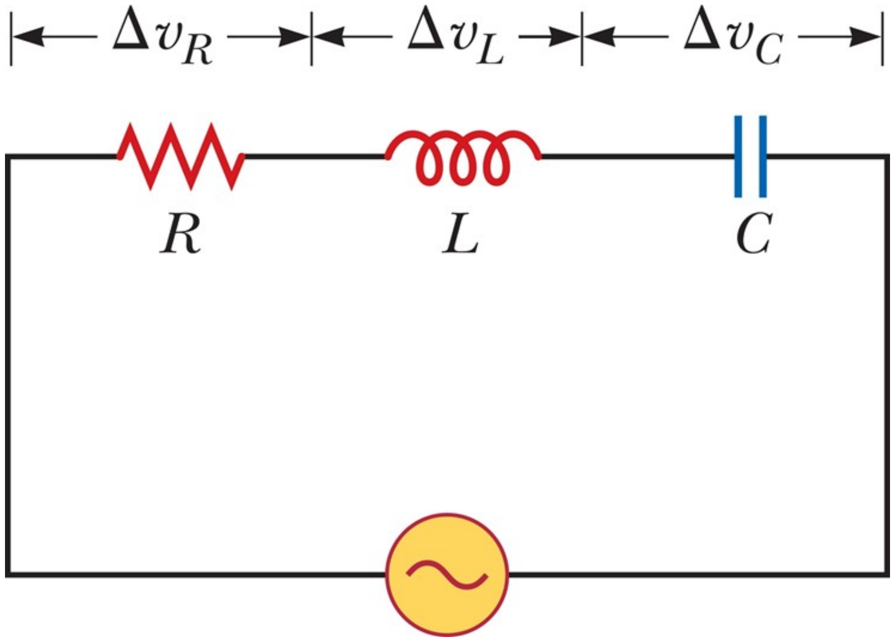
Inductors in an AC Circuit (2 of 2)



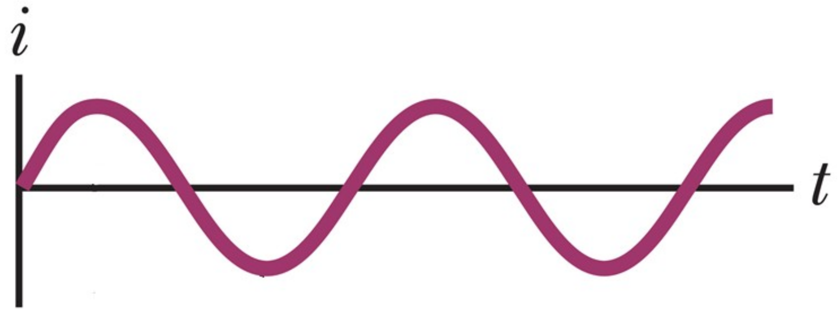
$$\Delta v_L = L\frac{\Delta I}{\Delta t}$$

The voltage across an inductor leads the current by 90°

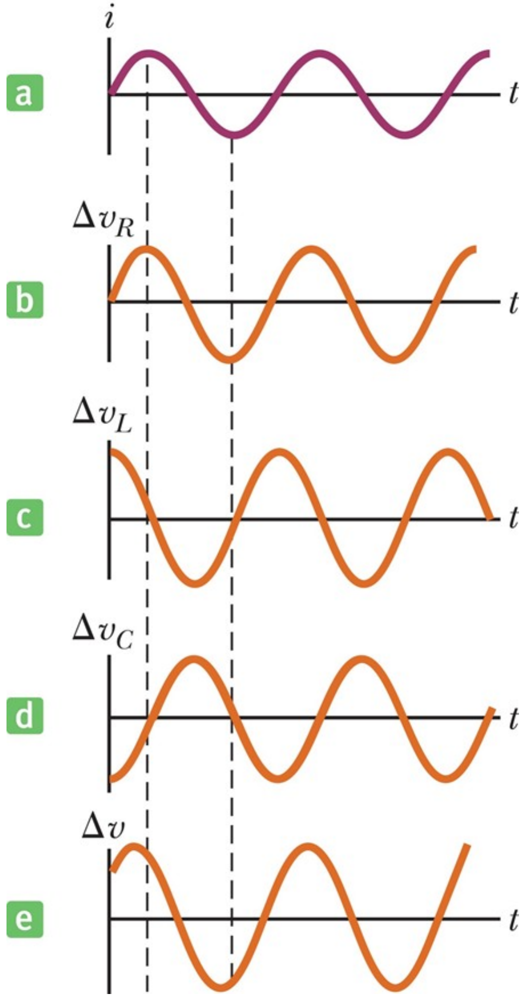
The *RLC* Series Circuit (1 of 7)



$$i = I_{max} \sin 2\pi f t$$

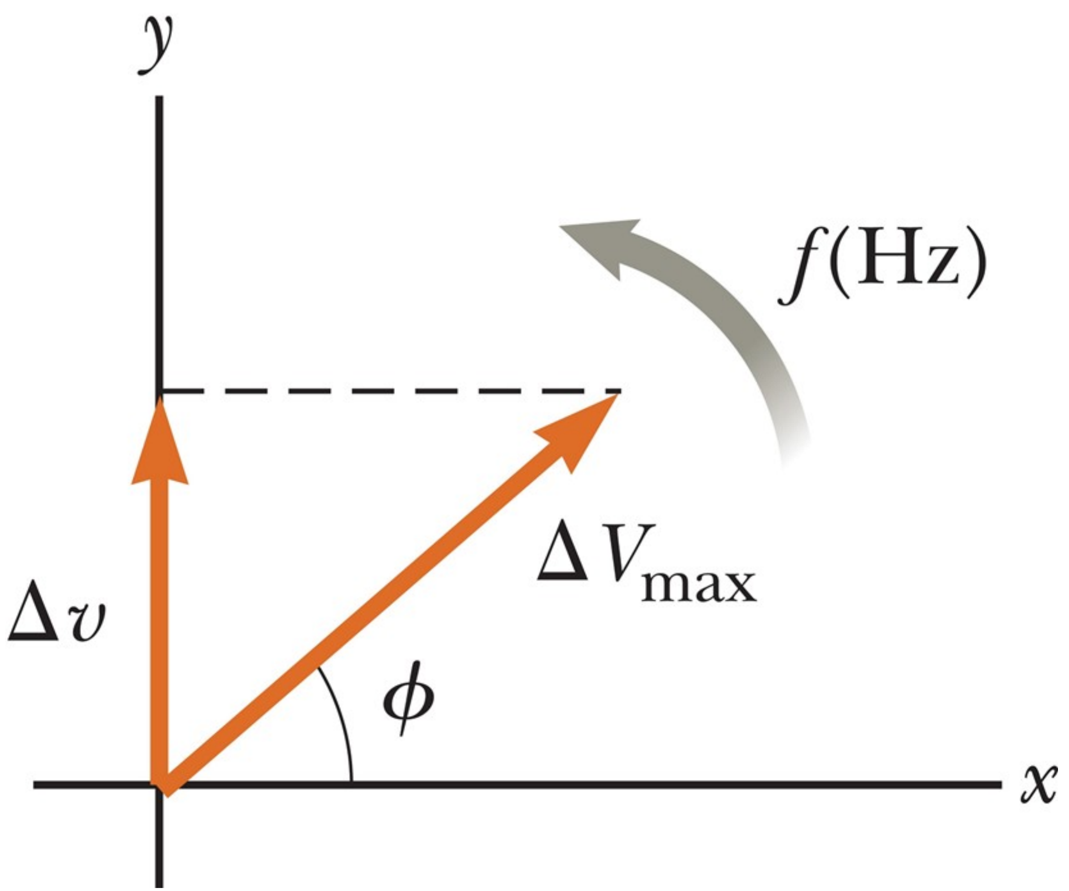


The *RLC* Series Circuit (2 of 7)



$$\Delta v = \Delta v_R + \Delta v_C + \Delta v_L$$

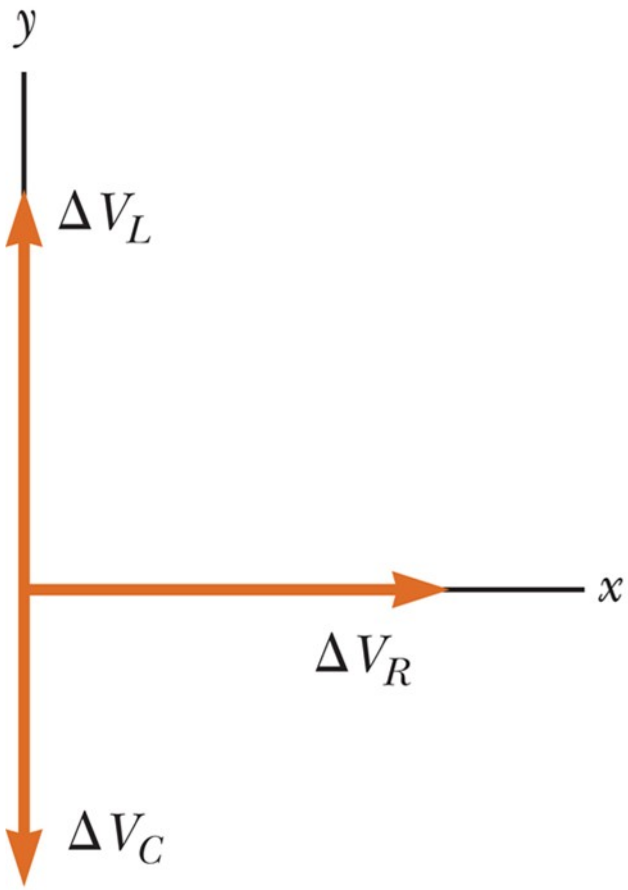
The *RLC* Series Circuit (3 of 7)

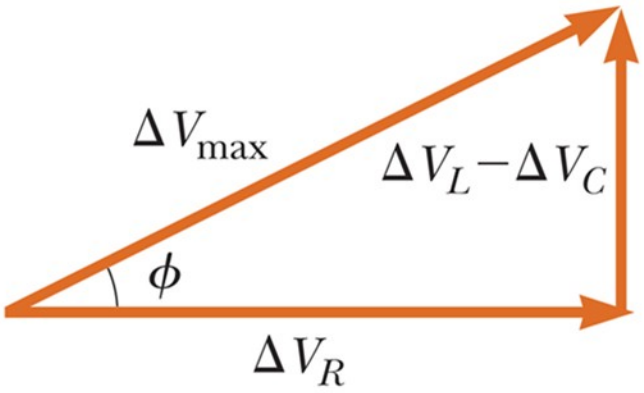


$$\Delta v = \Delta V_{max} \sin(2\pi f t + \phi)$$

$$i = i_{max} \sin(2\pi f t)$$

The *RLC* Series Circuit (4 of 7)





$$\Delta V_{max} = \sqrt{\Delta V_R^2 + (\Delta V_L - \Delta V_C)^2}$$

$$\tan\phi = \frac{\Delta V_L - \Delta V_C}{\Delta V_R}$$

The *RLC* Series Circuit

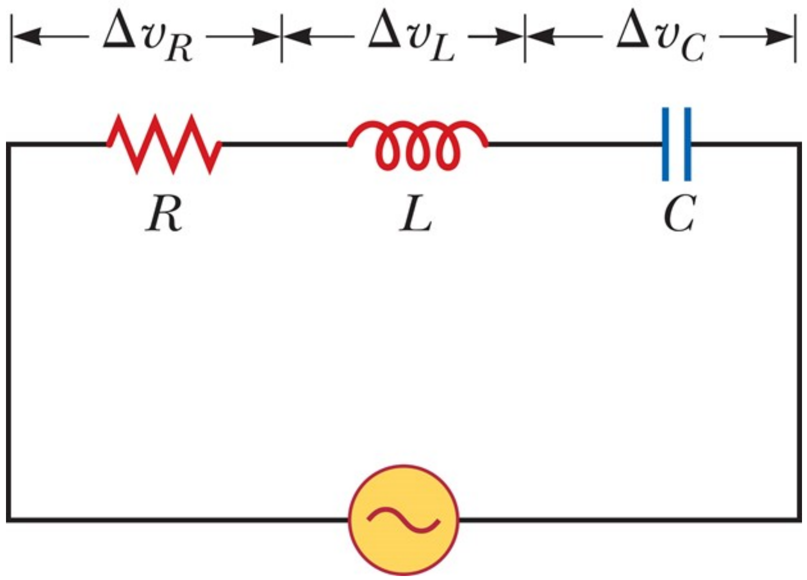
ELI the *ICE* man

- *ELI*: In an inductive circuit, the voltage leads the current.
- *ICE*: In a capacitive circuit the current leads the voltage.

Think – Pair – Share 2

For the circuit in the figure below, is the instantaneous voltage of the source is equal to

1. The sum of the maximum voltages across the elements.
2. The sum of the instantaneous voltages across the elements.
3. The sum of the rms voltages across the elements.



The *RLC* Series Circuit (6 of 7)

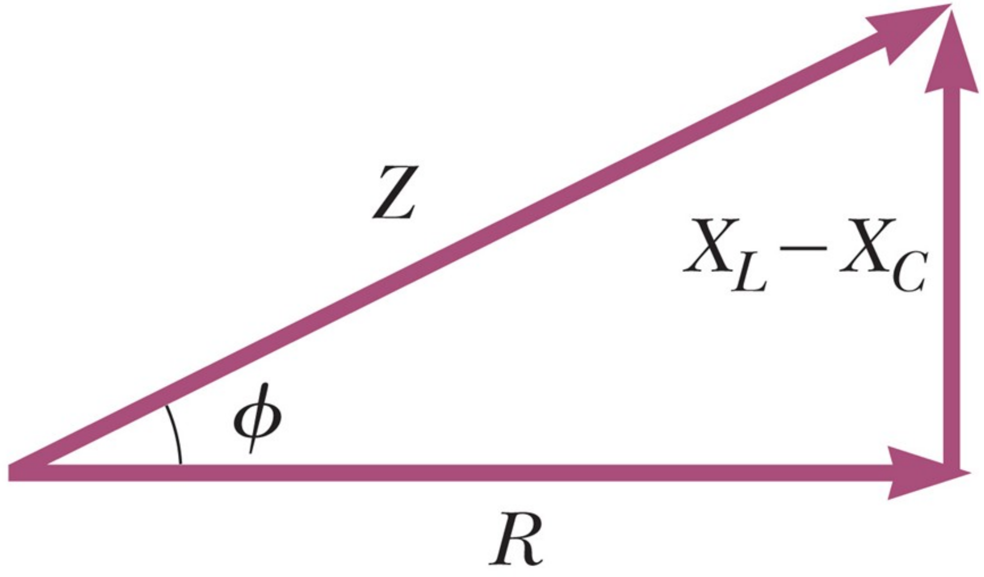
$$\Delta V_R = I_{\max} R, \quad \Delta V_L = I_{\max} X_L, \quad \text{and} \quad \Delta V_C = I_{\max} X_C$$

$$\Delta V_{\max} = I_{\max}\sqrt{R^2 + (X_L - X_C)^2}$$

$$Z \equiv \sqrt{R^2 + (X_L - X_C)^2}$$

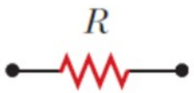
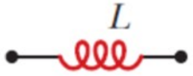
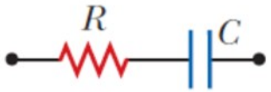
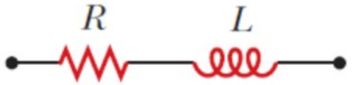
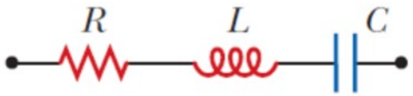
$$\Delta V_{\max} = I_{\max} Z$$

$$\tan\phi = \frac{X_L - X_C}{R}$$



The *RLC* Series Circuit

Table 21.2 Impedance Values and Phase Angles for Various Combinations of Circuit Elements

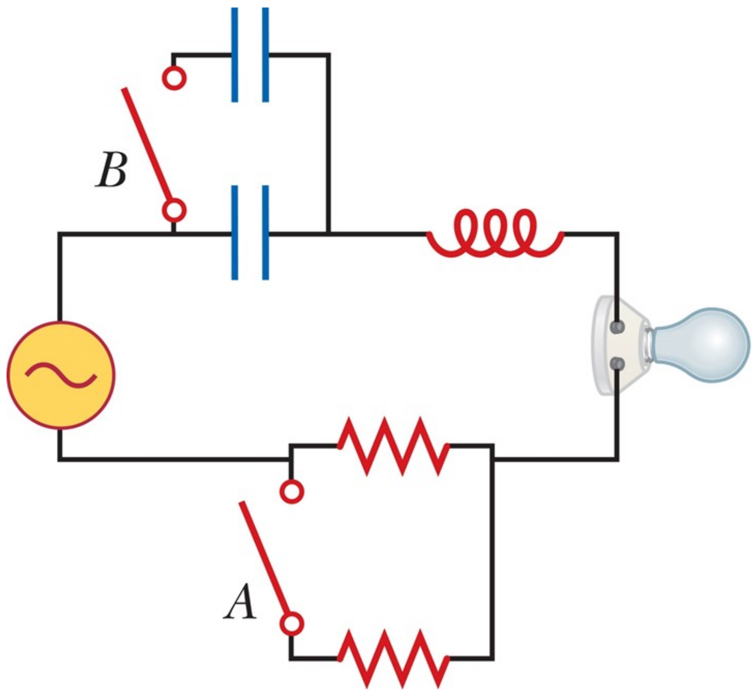
Circuit Elements	Impedance Z	Phase Angle ϕ
	R	0°
	X_C	−90°
	X_L	+90°
	$\sqrt{R^2 + X_C^2}$	Negative, between −90° and 0°
	$\sqrt{R^2 + X_L^2}$	Positive, between 0° and 90°
	$\sqrt{R^2 + (X_L - X_C)^2}$	Negative if $X_C > X_L$ Positive if $X_C < X_L$

Note: In each case, an AC voltage (not shown) is applied across the combination of elements (i.e., across the dots).

Think – Pair – Share 3

If switch A is closed in the figure below, what happens to the impedance of the circuit?

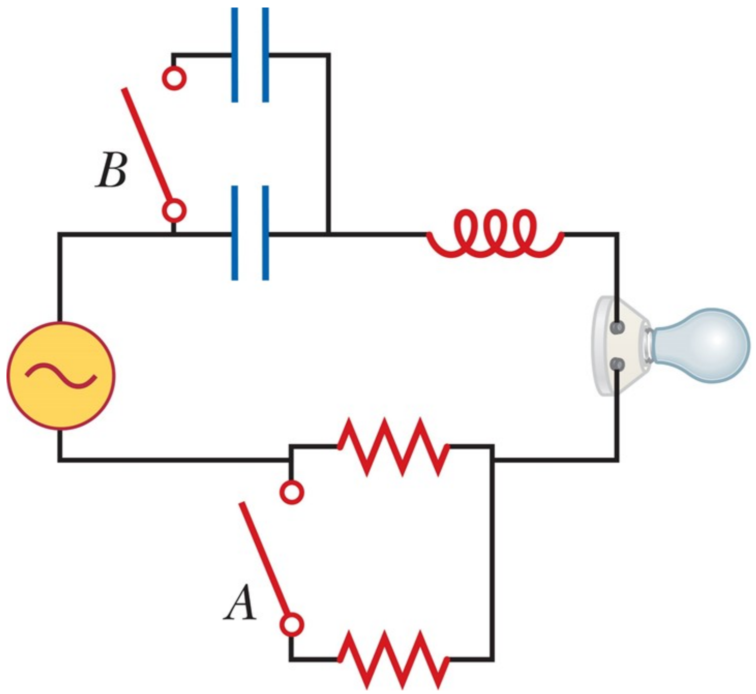
1. It increases.
2. It decreases.
3. It doesn't change.



Think – Pair – Share 4

Suppose $X_L > X_C$ in the figure. If switch A is closed, what happens to the phase angle?

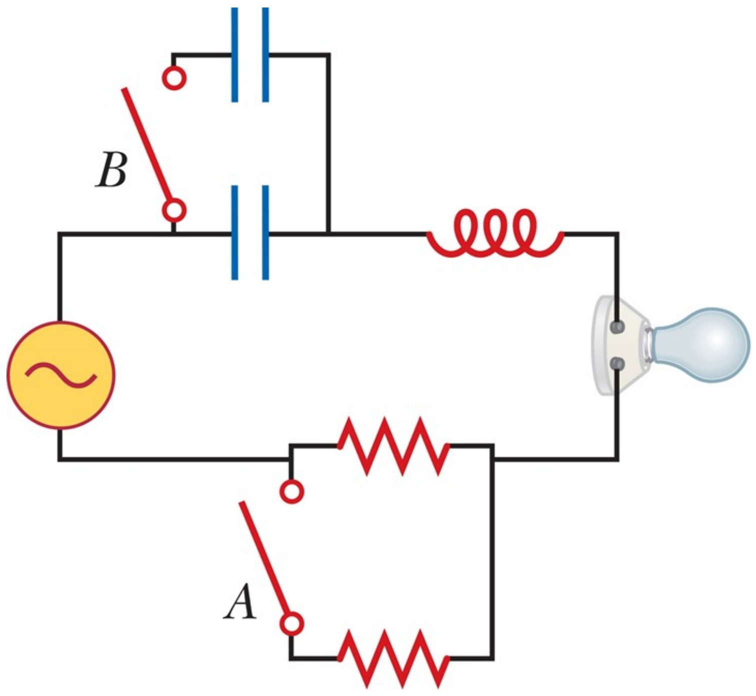
1. It increases.
2. It decreases.
3. It doesn't change.



Think – Pair – Share 5

Suppose $X_L > X_C$. If switch A is left open and switch B is closed in the figure below, what happens to the phase angle?

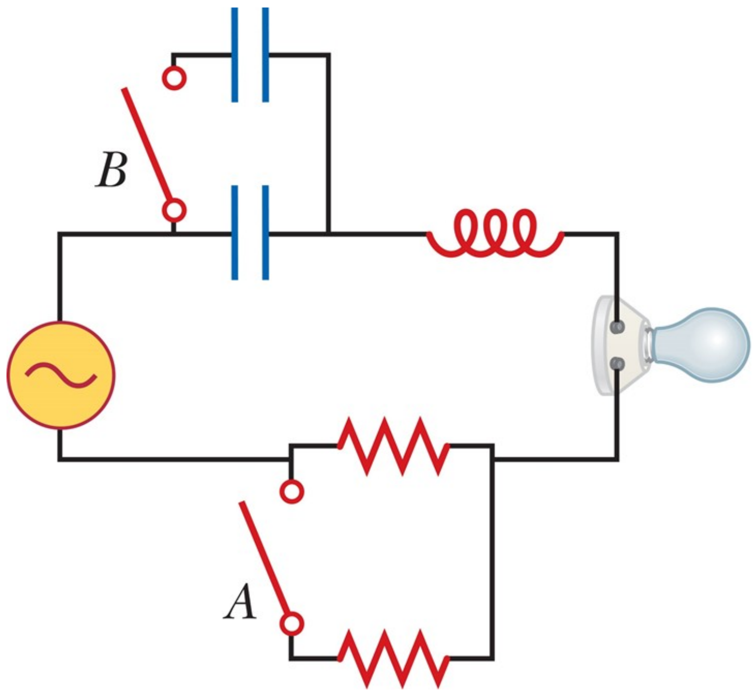
1. It increases.
2. It decreases.
3. It doesn't change.



Think – Pair – Share 6

Suppose $X_L > X_C$ in the figure below and, with both switches open, a piece of iron is slipped into the inductor. During this process, what happens to the brightness of the bulb?

1. It increases.
2. It decreases.
3. It doesn't change.



Problem-Solving Strategy: *RLC* Circuits

1. Calculate X_L and X_C.
2. Calculate the impedance Z
3. Use $\Delta V_{\max} = I_{\max} Z$
4. Calculate

$$\Delta V_R = I_{\max} R, \quad \Delta V_L = I_{\max} X_L, \quad \text{and} \quad \Delta V_C = I_{\max} X_C$$

5. Obtain the phase angle

$$\tan \phi = \frac{X_L - X_C}{R}$$

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Power in an AC Circuit (1 of 2)

$$PE_C = \tfrac{1}{2} C \left(\Delta V_{\text{max}}\right)^2$$

$$PE_L = \tfrac{1}{2} L I_{\text{max}}^2$$

$$P_{\text{av}} = I_{\text{rms}}^2 R$$

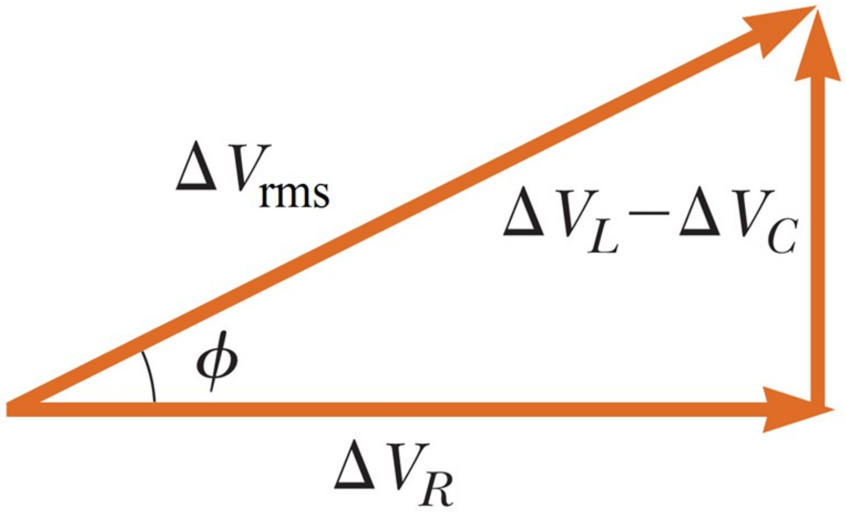
CENGAGE

Power in an AC Circuit (2 of 2)

$$P_{av} = I_{rms}^2 R \quad \rightarrow \quad R = \frac{\Delta V_{R,rms}}{I_{rms}} \quad \rightarrow \quad P_{av} = I_{rms}\,\Delta V_{R,rms}$$

$$\Delta V_{R,rms} = \Delta V_{rms}\cos\phi$$

$$P_{av} = I_{rms}\,\Delta V_{rms}\cos\phi$$

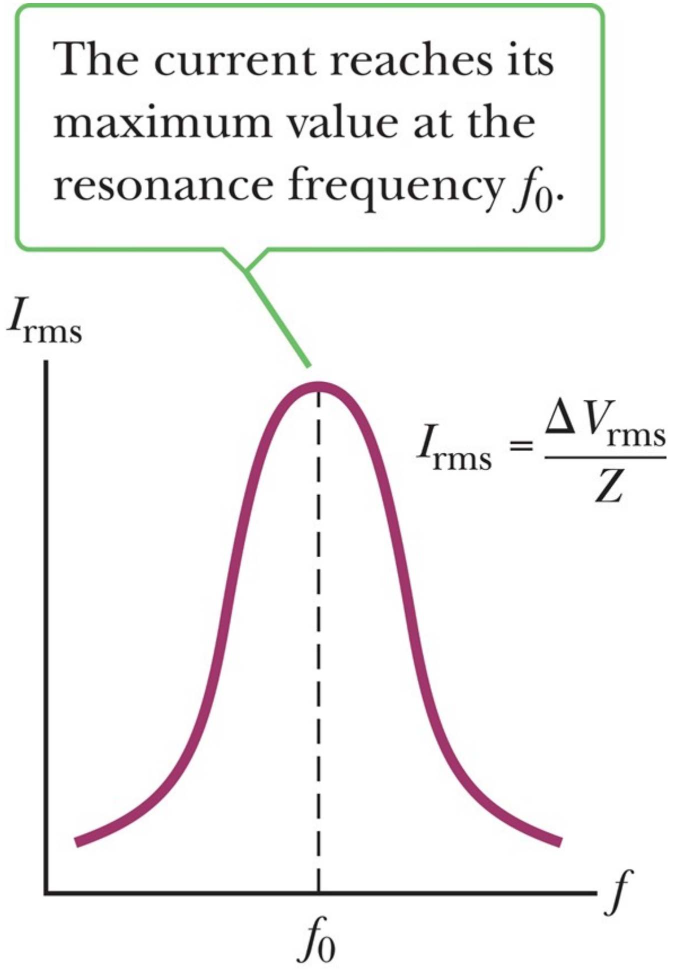


Resonance in a Series *RLC* Circuit (1 of 2)

$$I_{\text{rms}} = \frac{\Delta V_{\text{rms}}}{Z} = \frac{\Delta V_{\text{rms}}}{\sqrt{R^2 + (X_L - X_C)^2}}$$

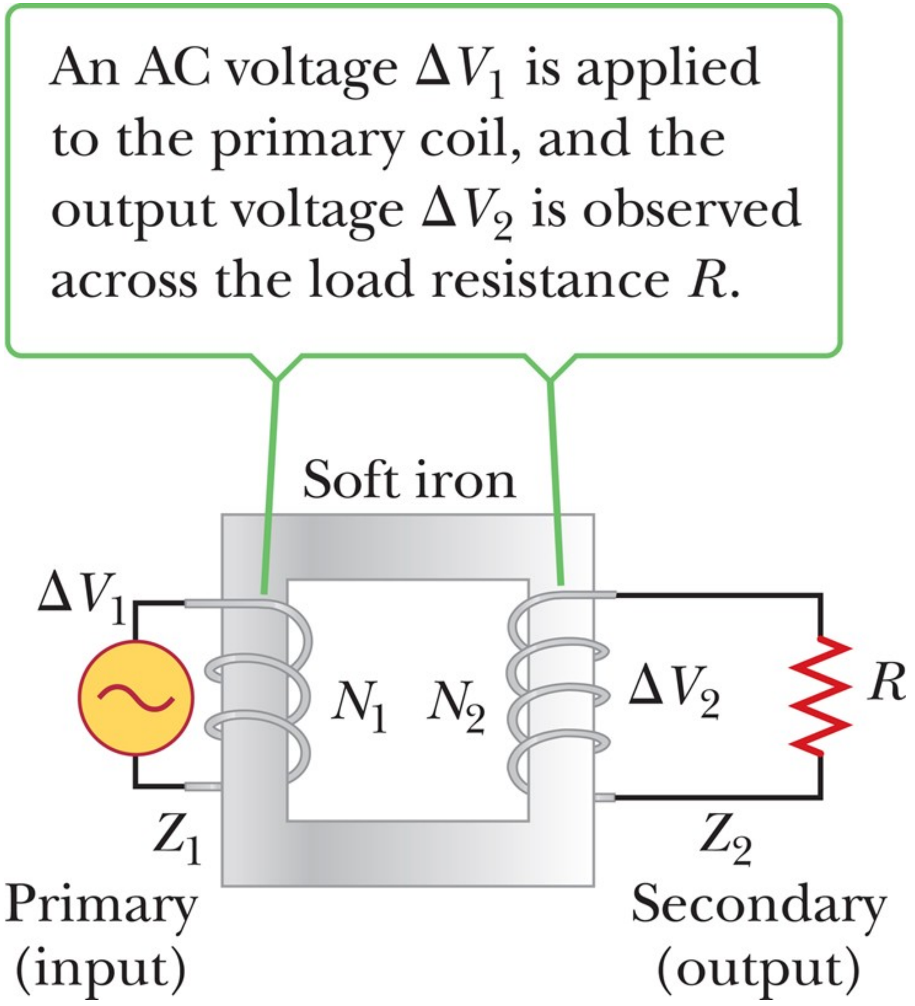
$$2\pi f_0 L = \frac{1}{2\pi f_0 C} \quad \rightarrow \quad f_0 = \frac{1}{2\pi\sqrt{LC}}$$

Resonance in a Series *RLC* Circuit (2 of 2)



$$I_{rms} = \frac{\Delta V_{rms}}{\sqrt{R^2 + (X_L - X_C)^2}}$$

The Transformer (1 of 3)

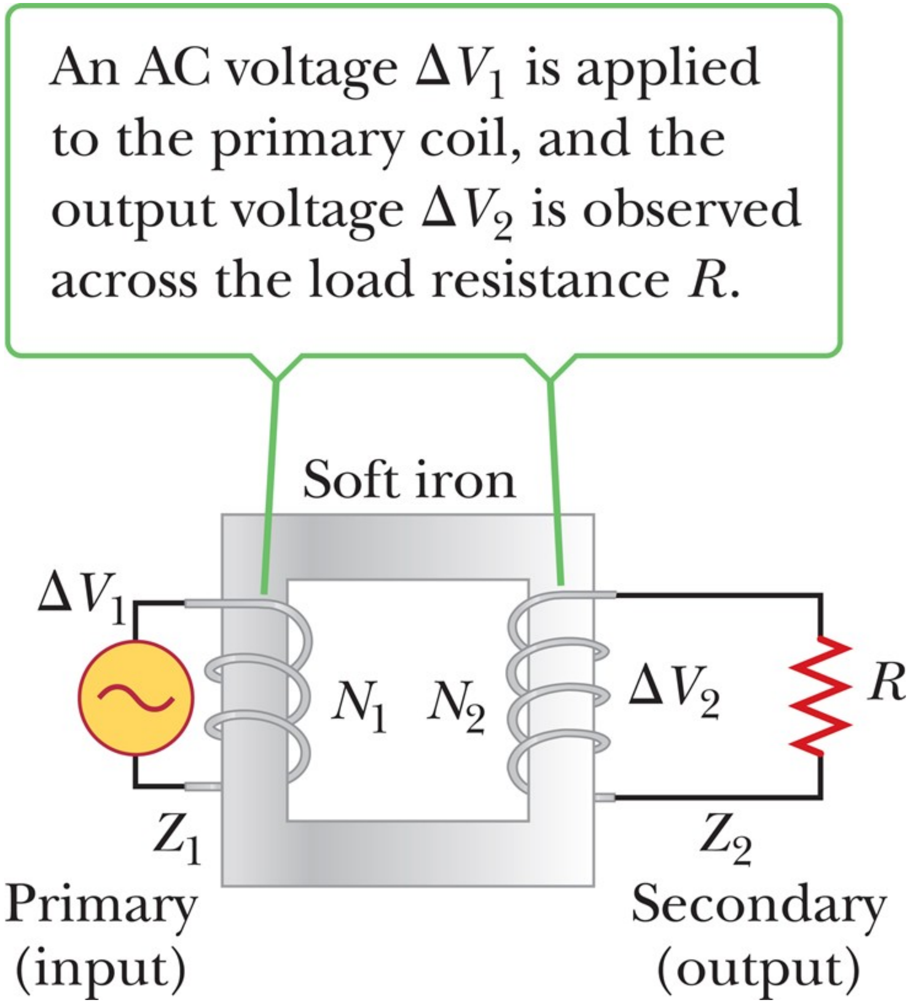


$$\Delta V_1 = -N_1 \frac{\Delta \Phi_B}{\Delta t}$$

$$\Delta V_2 = -N_2 \frac{\Delta \Phi_B}{\Delta t}$$

$$\Delta V_2 = -\frac{N_2}{N_1} \Delta V_1$$

The Transformer (2 of 3)



$$I_1 \Delta V_1 = I_2 \Delta V_2$$

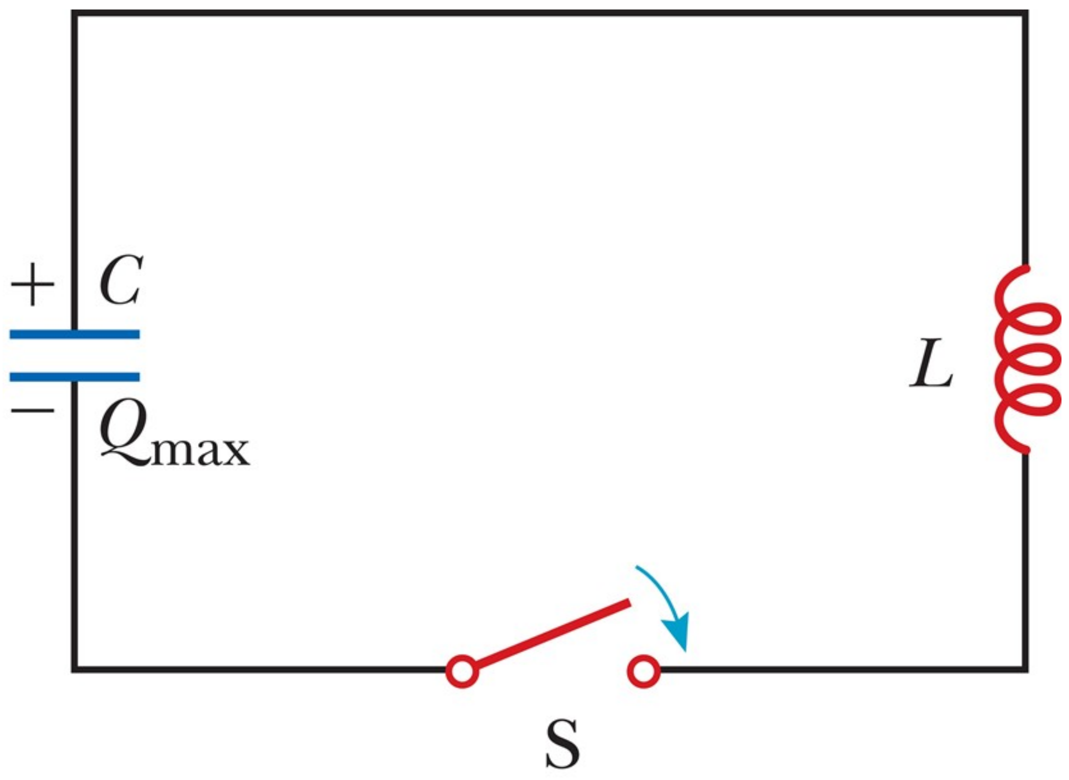
The Transformer (3 of 3)

Maxwell's Predictions (1 of 2)

1. Electric field lines originate on positive charges and terminate on negative charges.
2. Magnetic field lines always form closed loops.
3. A varying magnetic field induces an emf and so an electric field.
4. Magnetic fields are generated by moving charges or currents.

Maxwell’s Predictions (2 of 2)

Hertz's Confirmation of Maxwell's Predictions (1 of 3)

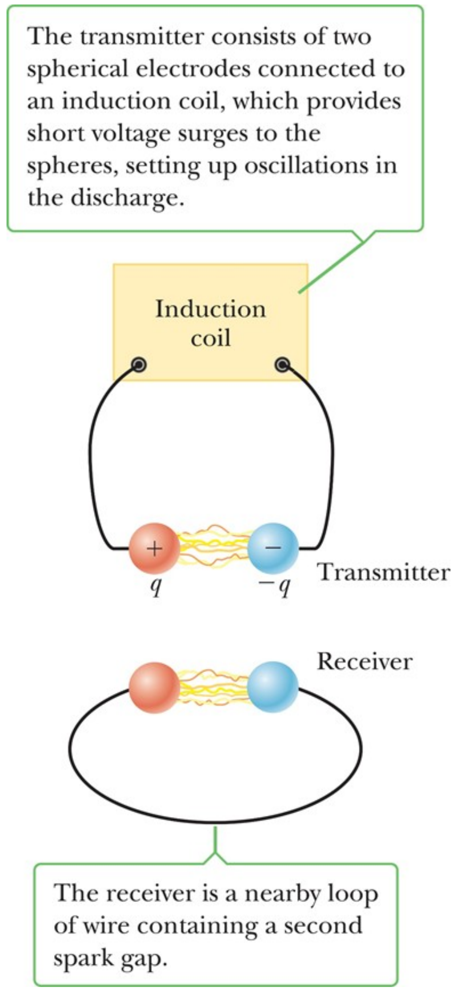


$$PE_C = \frac{Q_{\max}}{2C}$$

$$PE_L = \frac{L\,I^2}{2}$$

Hertz’s Confirmation of Maxwell’s Predictions (2 of 3)

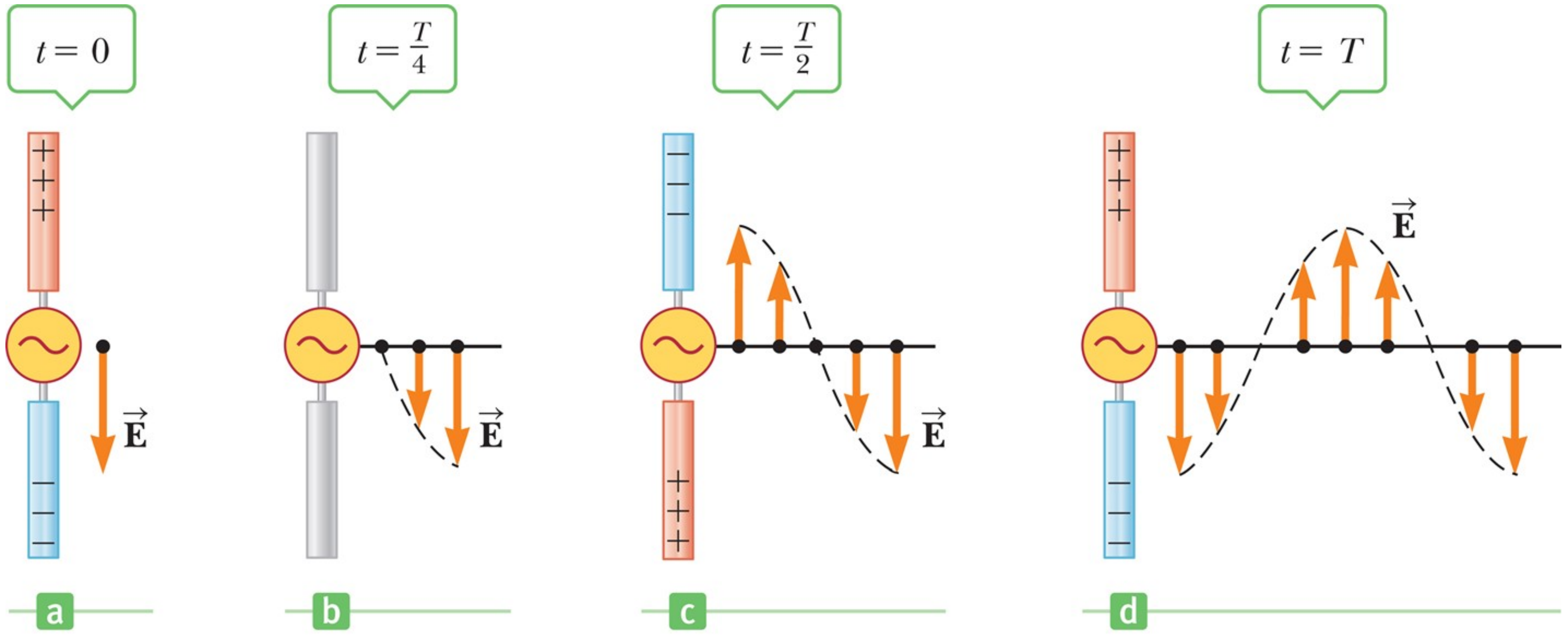
$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$



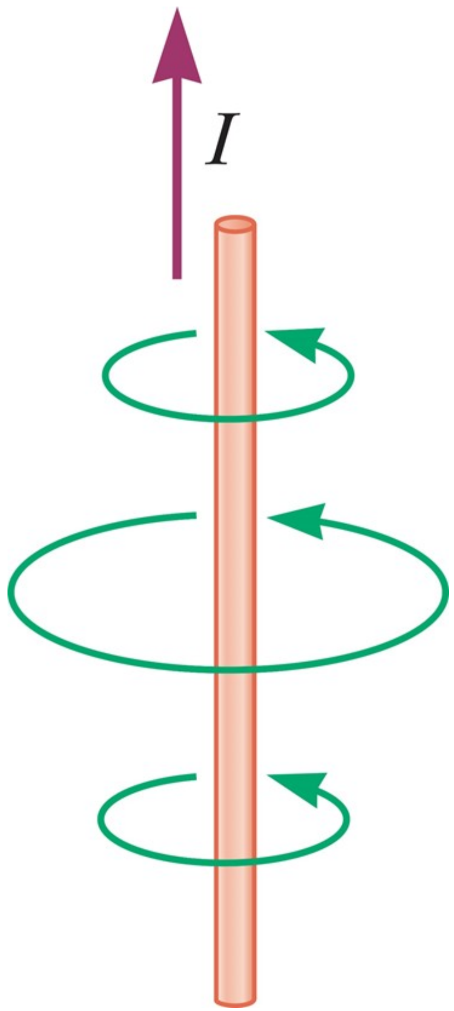
Hertz's Confirmation of Maxwell's Predictions (3 of 3)

$$v = \lambda f$$

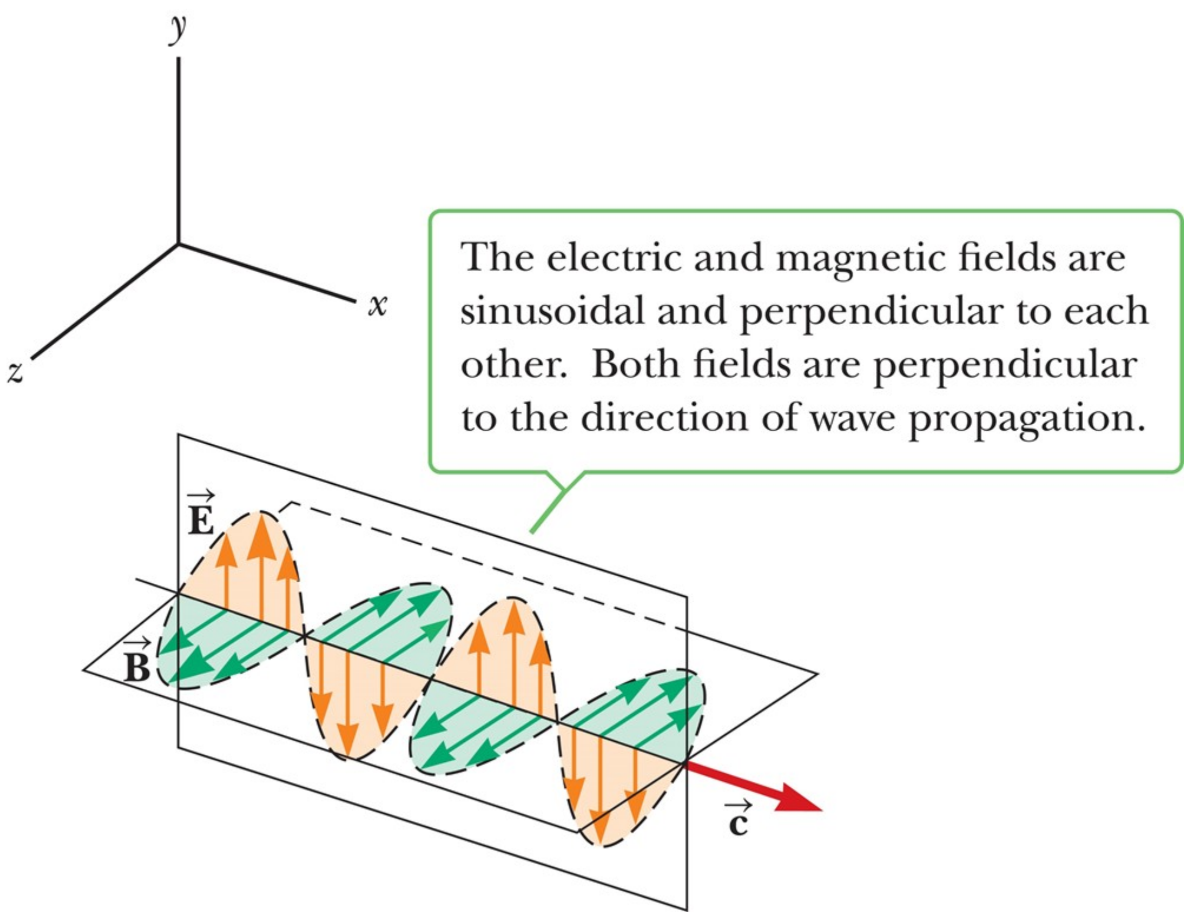
Production of Electromagnetic Waves by an Antenna (1 of 2)



Production of Electromagnetic Waves by an Antenna (2 of 2)



Properties of Electromagnetic Waves (1 of 6)



Properties of Electromagnetic Waves (2 of 6)

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$\mu_0 = 4\pi \times 10^{-7}\ \mathrm{N \cdot s^2/C^2}$$

$$\epsilon_0 = 8.854\,19 \times 10^{-12}\ \mathrm{C^2/(N \cdot m^2)}$$

$$c = 2.997\,92 \times 10^{8}\ \mathrm{m/s}$$

$$\frac{E}{B} = c$$

Properties of Electromagnetic Waves (3 of 6)

$$I = \frac{E_{\max} B_{\max}}{2\mu_0}$$

$$I = \frac{E_{\max}^2}{2\,\mu_0\,c} = \frac{c}{2\mu_0} B_{\max}^2$$

$$\frac{E}{B} = c \quad \rightarrow \quad E_{\max} = c\,B_{\max} = \frac{B_{\max}}{\sqrt{\mu_0 \epsilon_0}}$$

Properties of Electromagnetic Waves (4 of 6)

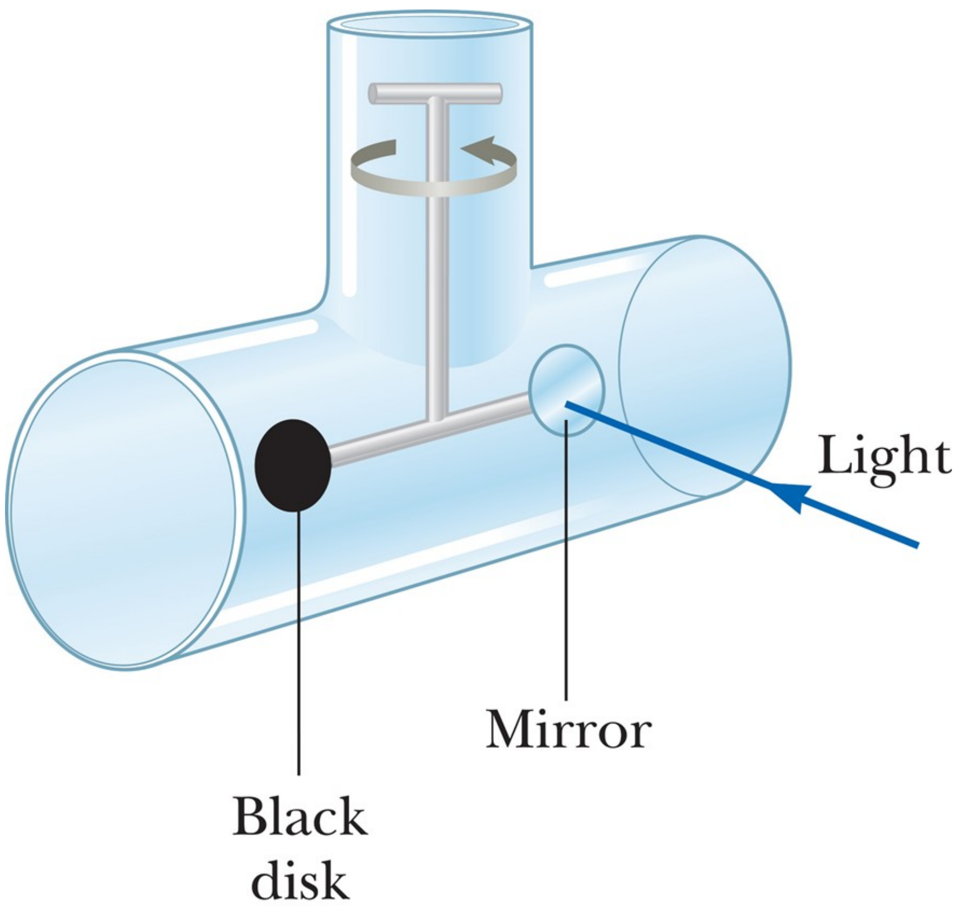
$$U = IA\Delta t$$

$$p = \frac{U}{c} \qquad \text{(complete absorption)}$$

$$\Delta p = mv - (-mv) = 2mv$$

$$p = \frac{2U}{c} \qquad \text{(complete reflection)}$$

Properties of Electromagnetic Waves (5 of 6)



Properties of Electromagnetic Waves (6 of 6)

1. EM waves travel at *c*
2. EM waves are transverse waves
3. $\dfrac{E}{B} = c$
4. Electromagnetic waves carry both energy and momentum

Think – Pair – Share 7

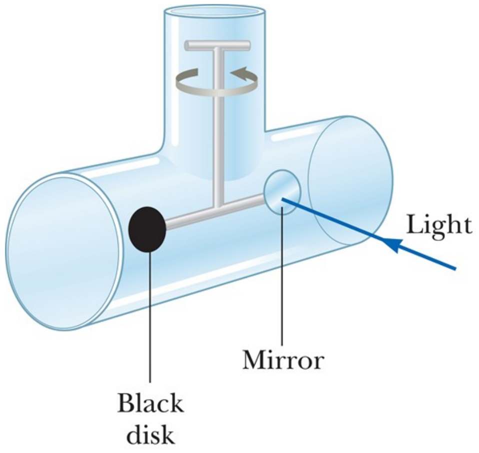
In an apparatus such as that in the figure below, suppose the black disk is replaced by one with half the radius. Which of the following are different after the disk is replaced?

a) radiation pressure on the disk

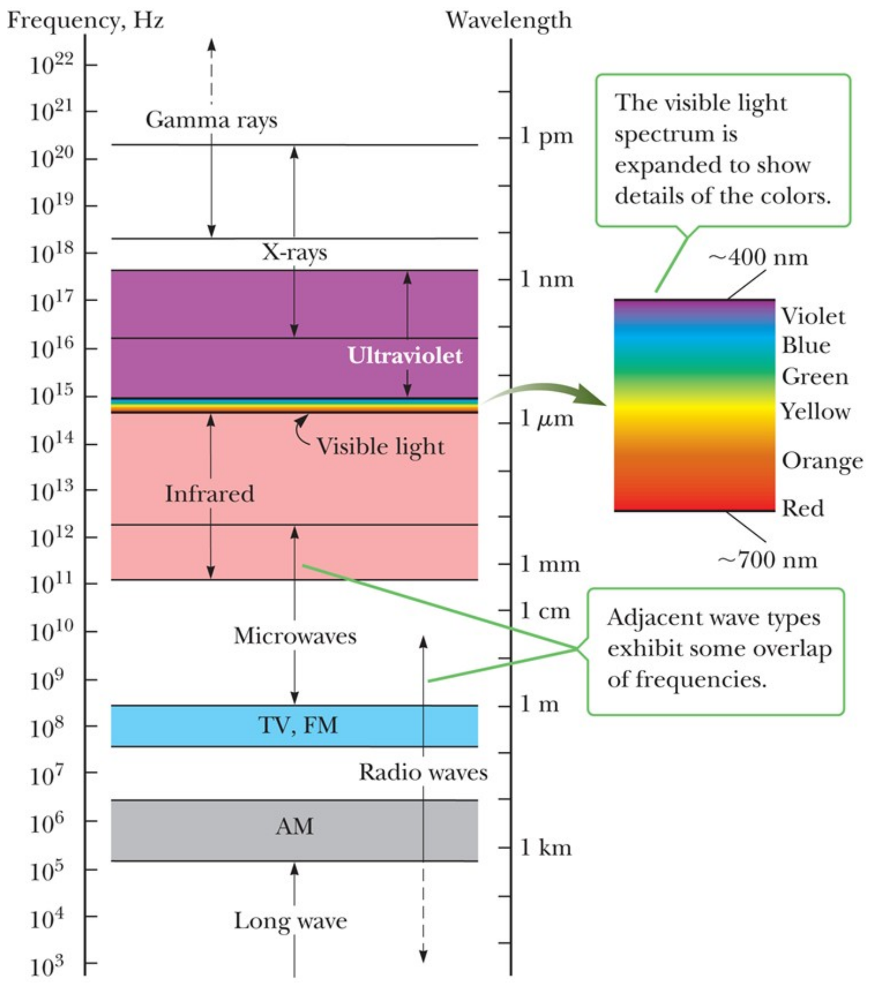
b) radiation force on the disk

c) radiation momentum delivered to the disk in a given time interval

1. (a) only
2. (b) and (c)
3. (a) (b) and (c)
4. (b) only



The Spectrum of Electromagnetic Waves (1 of 10)



The Spectrum of Electromagnetic Waves

$$1 \text{ micrometer } (\mu\text{m}) = 10^{-6} \text{ m}$$

$$1 \text{ nanometer } (\text{nm}) = 10^{-9} \text{ m}$$

$$1 \text{ angstrom } (\text{Å}) = 10^{-10} \text{ m}$$

Think – Pair – Share 8

Which of the following statements are true about light waves?

a) The higher the frequency, the longer the wavelength.

b) The lower the frequency, the longer the wavelength.

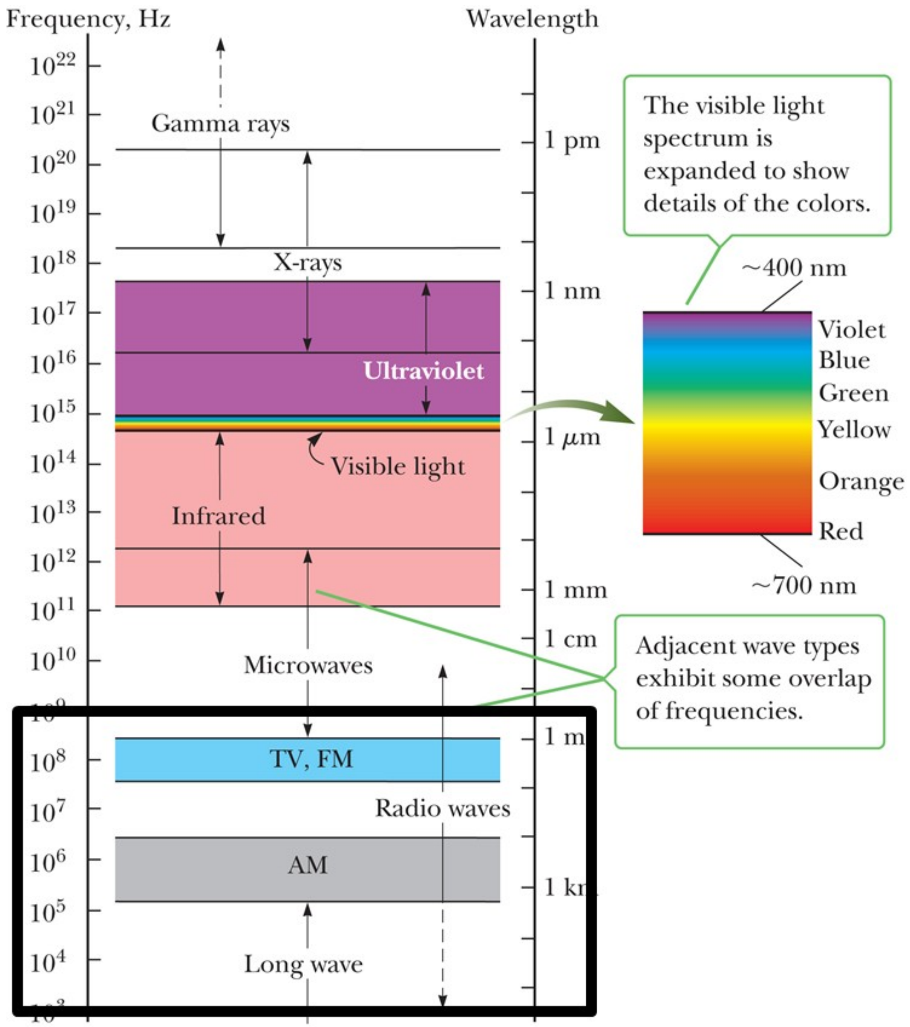
c) Higher-frequency light travels faster than lower-frequency light.

d) The shorter the wavelength, the higher the frequency.

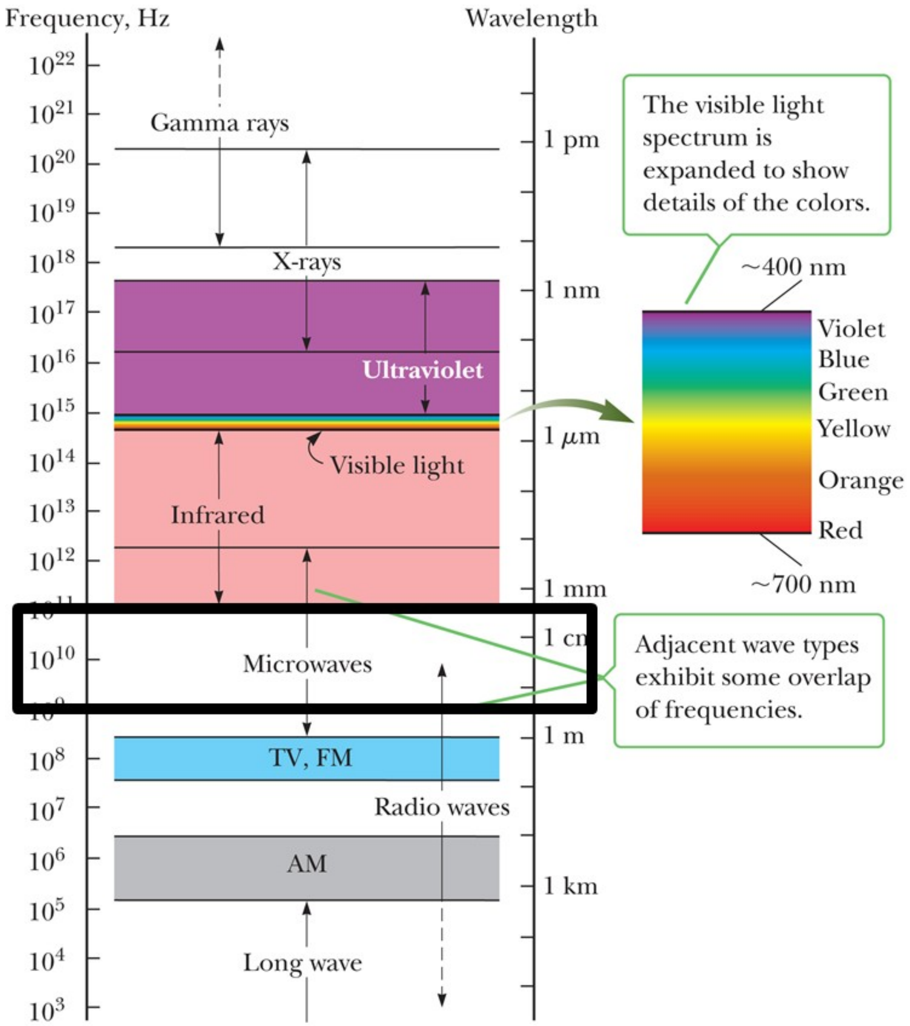
e) The lower the frequency, the shorter the wavelength.

1. (b) only
2. (c) and (d)
3. (b) and (d)
4. (b) only

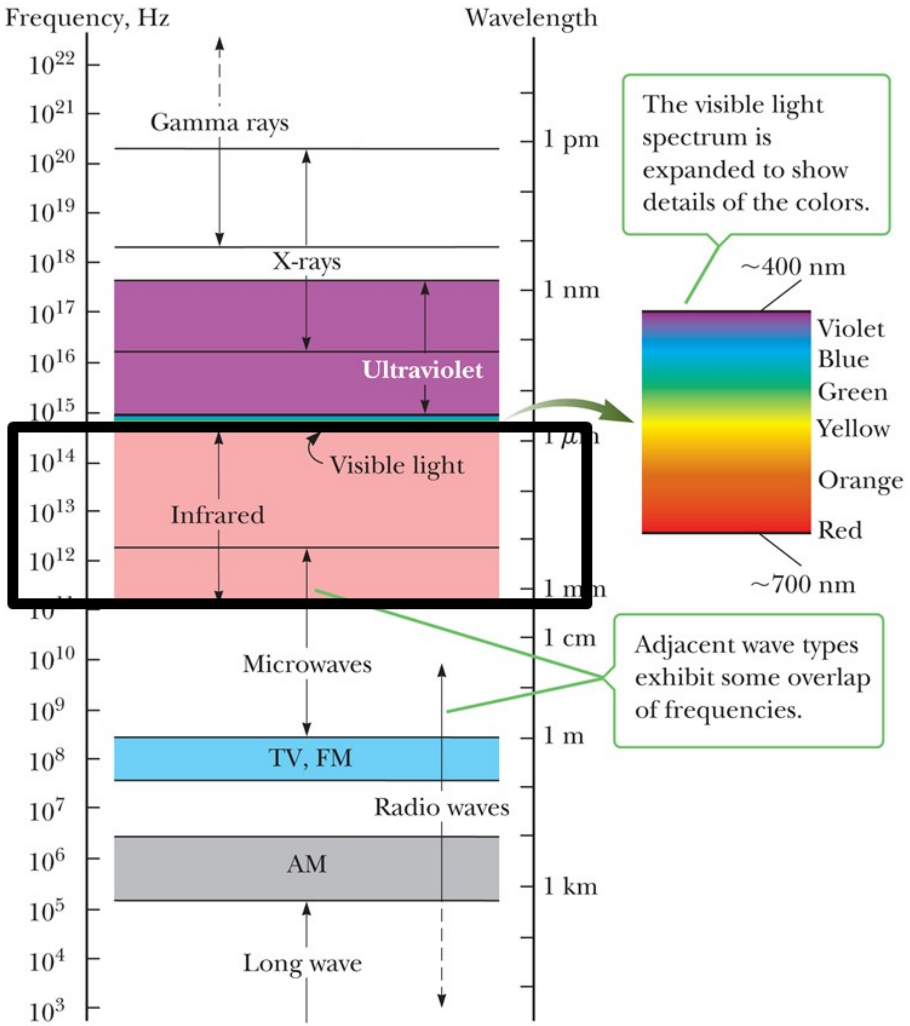
The Spectrum of Electromagnetic Waves (3 of 10)



The Spectrum of Electromagnetic Waves (4 of 10)



The Spectrum of Electromagnetic Waves (5 of 10)



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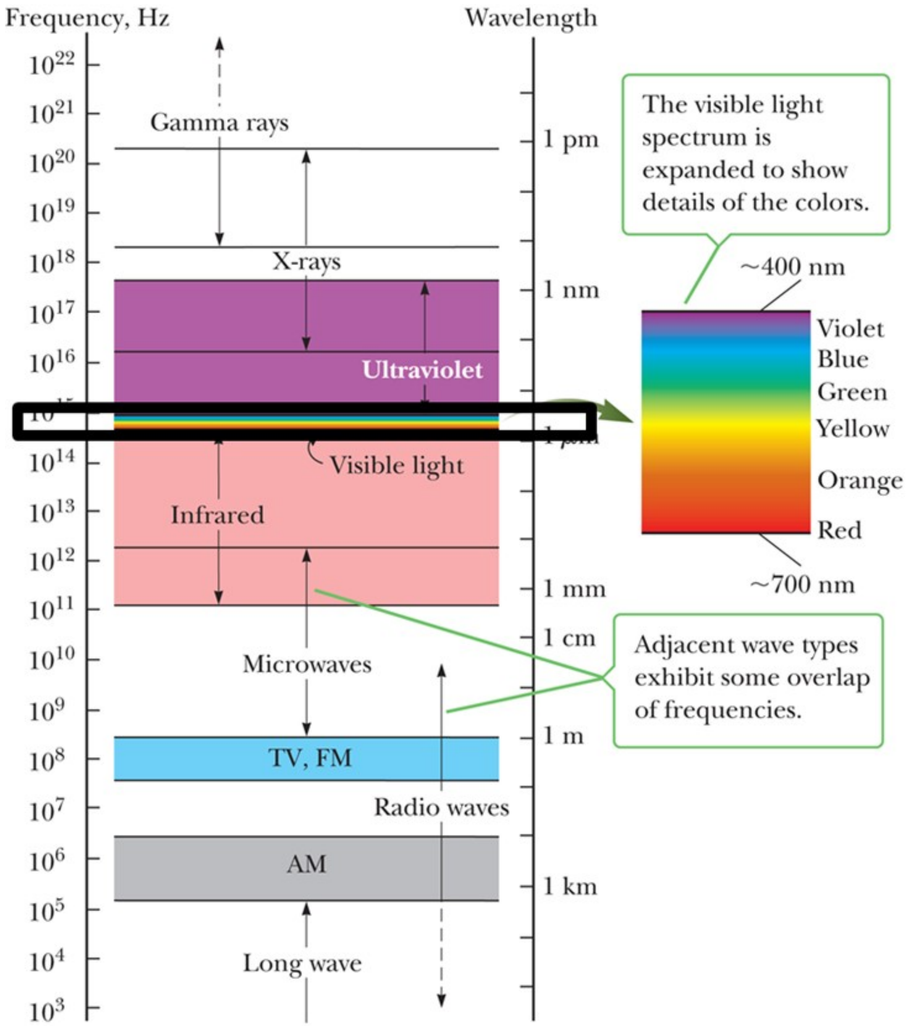
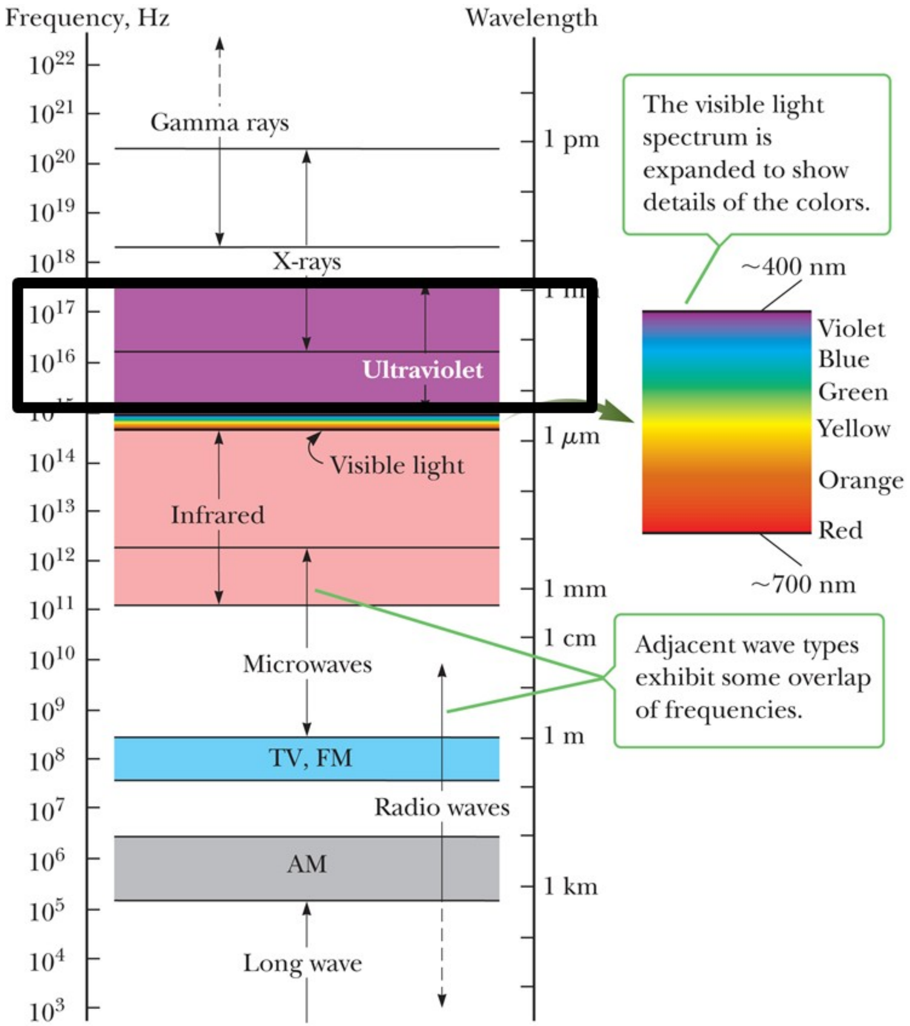
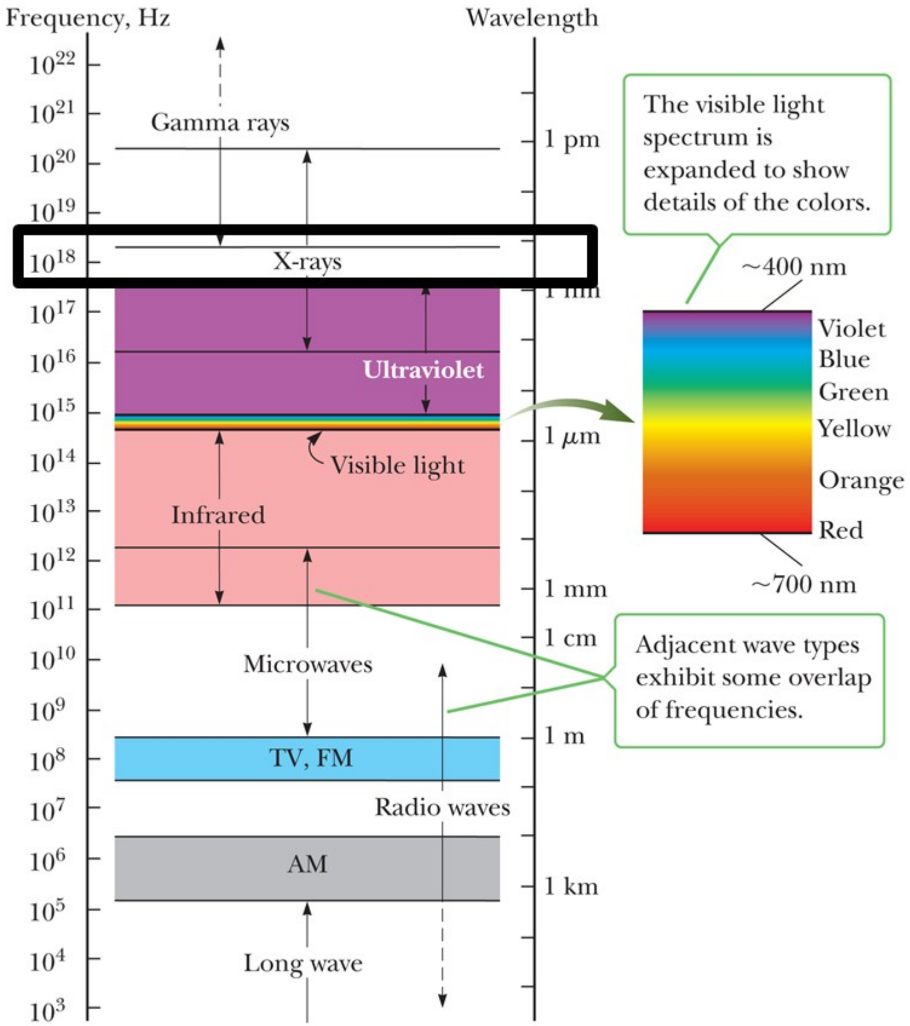
The Spectrum of Electromagnetic Waves (6 of 10)



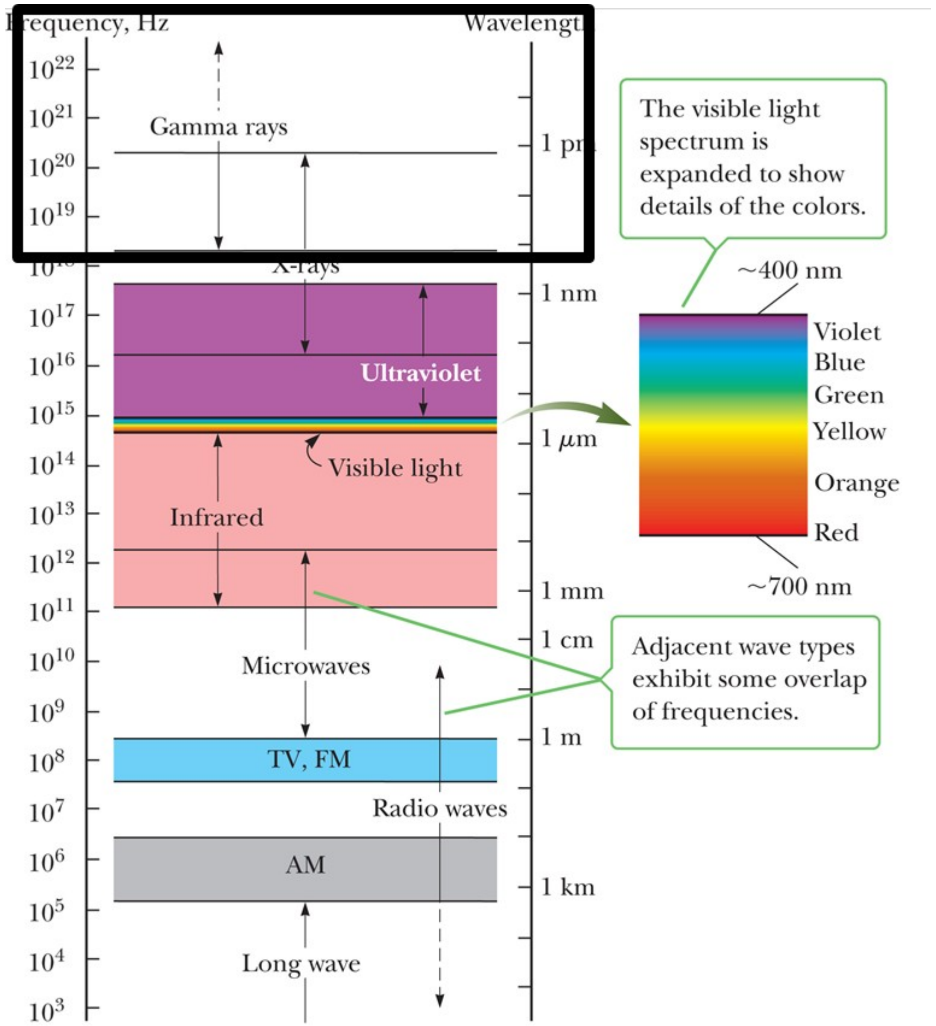
The Spectrum of Electromagnetic Waves (7 of 10)



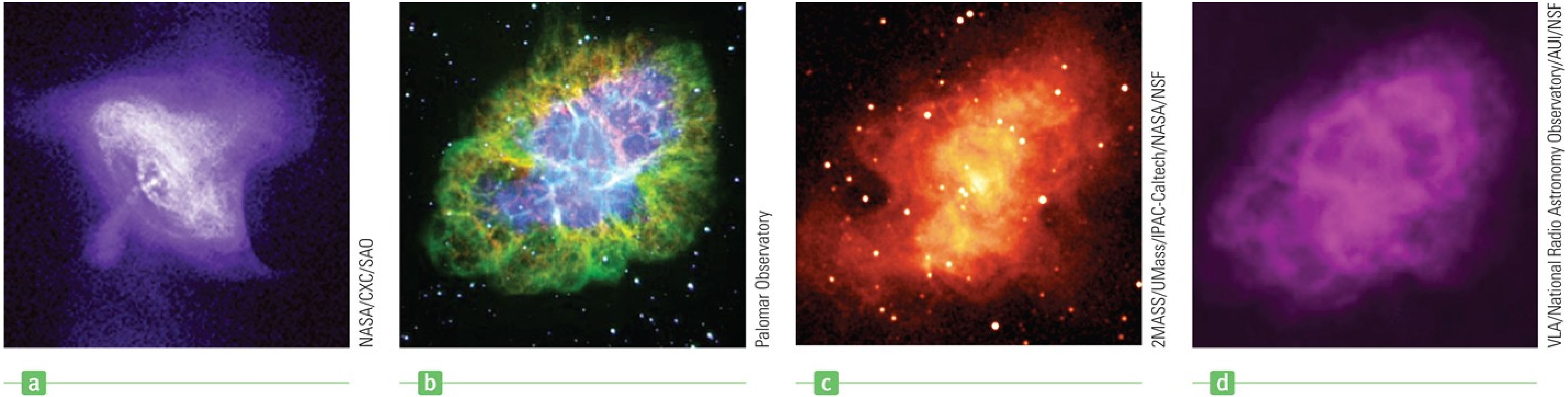
The Spectrum of Electromagnetic Waves (8 of 10)



The Spectrum of Electromagnetic Waves (9 of 10)



The Spectrum of Electromagnetic Waves (10 of 10)



The Doppler Effect for Electromagnetic Waves (1 of 2)

- Sound waves require a medium, EM waves do not.
- Speed of sound in Doppler effect equation for sound depends on the reference frame, the speed of EM waves has the same value in all coordinate systems at rest or moving at constant velocity with respect to one another.

$$f_0 \approx f_S\left(1 \pm \frac{u}{c}\right) \quad \text{if } u << c$$

The Doppler Effect for Electromagnetic Waves (2 of 2)